

Take-Home Exercise: Pre-Op Scheduling Triage

Overview

We are excited to see your creativity and skills in action! For this exercise, you are building a clinical triage agent for Cadence Surgical Center.

Surgeries cannot be scheduled until required documentation, testing, and risk checks are complete. You will build a system that evaluates a patient's pre-op state and determine:

1. Whether the patient can be scheduled
2. What items are missing or blocking

This is **not a medical knowledge test** (most engineers did not have prior healthcare background before working at Cadence!). Instead, we're looking for strong reasoning and system design. Do not use external medical guidelines and use the policy below (see Appendix) as the only source of truth.

Input

You will receive a single patient submission package as JSON.

The structure resembles what might be returned from a healthcare API and may include:

- Patient demographics
- Procedure information
- Vital signs
- Laboratory results
- Active medications
- Clinical notes or documents

Important information may appear in free text documents. You should make reasonable assumptions and document them.

Output

Your system must return valid JSON in the following format. Every issue must reference the exact field value, date, or document excerpt used. Example output below:

JSON

```
{
  "decision": "NEEDS_FOLLOW_UP",
  "issues": [
    {
      "category": "MISSING_REQUIRED_DATA",
      "description": "Missing procedure date",
      "evidence": {
        "source": "procedure.procedure_date",
        "details": "procedure.procedure_date is null"
      }
    },
    {
      "category": "ANTICOAGULATION_MANAGEMENT",
      "description": "Missing perioperative anticoagulation plan",
      "evidence": {
        "source": "documents[4]",
        "details": "Active anticoagulant medication present (medications[1]) but no clear perioperative plan document found"
      }
    }
  ],
  "explanation": "MISSING_REQUIRED_DATA: Missing procedure date | ANTICOAGULATION_MANAGEMENT: Missing perioperative anticoagulation plan"
}
```

Submission Guidelines

This [repo](#) includes a naive baseline Python implementation that makes a single LLM call as well as a dataset, harness, and report viewer for iteration. A perfect score on the sample data matters less than how you approached the problem. We're looking for strong engineering judgment — evidence that you care about architecture, craft, and outcomes.

You may use any language and any libraries. You are welcome to use AI to help with any part of the assignment and as part of the solution. We are happy to share an OpenAI API Key.

When sharing your solution, please include:

- Instructions on how to build and execute if they are different from the starter code
- A brief write-up of your approach and design decisions
- An export of your coding agent session(s):
 - For Claude, use `/export`
 - For Codex, attach the relevant jsonl files in `~/ .codex/sessions/`

Appendix

Cadence Surgical Center Pre-Operative Scheduling Policy

Effective: January 1, 2026

Purpose

To establish standardized clinical requirements for determining patient readiness for surgical scheduling at Cadence Surgical Center.

Patient Clearance Status

Each patient must be assigned **exactly one** of the following pre-operative clearance statuses:

- **READY** – All required documentation, testing, and clinical criteria are complete and within defined timeframes.
- **NEEDS_FOLLOW_UP** – Required documentation, testing, or perioperative planning is missing, incomplete, or outside required timeframes.
- **NOT_CLEARED** – The patient has a safety condition that precludes proceeding with the surgery.

Rule 1: Required Documentation

The following documentation must be present prior to scheduling:

1. **History and Physical (H&P)**
 - Must be completed within **30 days** of the scheduled procedure date.
2. **Signed Surgical Consent**

If any required documentation is missing or outdated → **NEEDS_FOLLOW_UP**

Rule 2: Required Pre-Operative Testing by Procedure Risk Level

Each procedure must be categorized as **LOW**, **MODERATE**, or **HIGH** risk.

2.1 LOW or MODERATE Risk Procedures

- **Complete Blood Count (CBC)** within **30 days** of the procedure date.

2.2 HIGH Risk Procedures

- **Complete Blood Count (CBC)** within **14 days**
- **Comprehensive Metabolic Panel (CMP)** within **14 days**

Testing Review Standards

- Only the most recent result for each required test shall be considered.
- Results must fall within the specified timeframe relative to the procedure date.
- If any required test is missing or outside the permitted timeframe, classify the patient as **NEEDS_FOLLOW_UP**

Rule 3: Anticoagulation Management

For patients currently taking anticoagulant medication:

- A documented **perioperative anticoagulation management plan** is required prior to scheduling.
- The plan must clearly describe how the medication will be managed before and after the procedure.

If there is no documented plan, or the plan is incomplete or ambiguous, classify the patient as **NEEDS_FOLLOW_UP**.

Rule 4: Acute Safety Exclusion Criteria

The presence of any of the following conditions at the time of pre-operative review will result in **NOT_CLEARED** status:

- Systolic blood pressure ≥ 180 mmHg
- Diastolic blood pressure ≥ 110 mmHg
- Temperature $> 100.4^{\circ}\text{F}$

Only the most recent result for each required test shall be considered.

Final Clearance Determination

A patient may be classified as **READY** only when:

- All required documentation is present and current
- All required testing is complete and within defined timeframes
- Anticoagulation management (if applicable) is clearly documented
- No acute safety exclusion criteria are present

If a required field needed to evaluate a rule is missing/unknown, the patient should be classified as **NEEDS_FOLLOW_UP**.

Patient Submission Format

Example Input:

JSON

```
{
  "patient": {
    "id": "14401b46-16cf-5c04-b43a-fec3e8485985",
    "mrn": "MRN-3000000",
    "name": {
      "given": "Danielle",
      "family": "Lopez"
    },
    "dob": "1946-01-01",
    "sex": "F"
  },
  "procedure": {
    "case_id": "case_00000",
    "procedure_type": "Elective total knee arthroplasty",
    "procedure_risk": "MODERATE",
    "procedure_date": null,
    "is_elective": true,
    "location": "Cadence Surgical Center - Main OR"
  },
  "vitals": [
    {
      "type": "blood_pressure",
      "systolic": 117,
      "diastolic": 72,
      "date": "2026-02-17T09:05:00Z",
      "source": "Pre-op clinic"
    },
    {
      "type": "blood_pressure",
      "systolic": 122,
      "diastolic": 76,
      "date": "2026-02-24T10:12:00Z",
      "source": "Primary care visit"
    },
    {
      "type": "temperature",
      "value_f": 98.8,
      "date": "2026-02-24T10:15:00Z",
      "source": "Pre-op clinic"
    }
  ],
  "labs": [
    {
      "id": "69d20291-f423-54f9-9d5c-14b6608351c4",
      "code": "CBC",
      "display": "Complete Blood Count",
      "effective_at": "2026-02-21T08:10:00Z",
    }
  ]
}
```

```

    "status": "final",
    "source": "Internal Hospital Lab"
  },
  {
    "id": "58de471c-f6b4-5567-9b57-819f93e57e67",
    "code": "CBC",
    "display": "CBC (Complete Blood Count)",
    "effective_at": "2026-02-03T08:25:00Z",
    "status": "final",
    "source": "Internal Hospital Lab"
  },
  {
    "id": "682bf9a8-3325-57bf-a231-5f8ad1185692",
    "code": "HBA1C",
    "display": "Hemoglobin A1c",
    "effective_at": "2026-02-09T07:40:00Z",
    "status": "final",
    "source": "Regional Reference Lab"
  }
],
"medications": [
  {
    "name": "lisinopril",
    "active": true
  },
  {
    "name": "apixaban",
    "active": true
  }
],
"conditions": [
  {
    "name": "Atrial fibrillation",
    "active": true
  }
],
"documents": [
  {
    "doc_id": "c92f23f5-bc1f-54cb-911f-c719bc7df37b",
    "type": "History and Physical",
    "date": "2026-02-19",
    "author": "Danielle Rivera, APRN",
    "text": "HISTORY AND PHYSICAL: pre-op evaluation complete for planned procedure.
Procedure date target: 2026-03-01."
  },
  {
    "doc_id": "bdb55172-db02-5721-a48e-81b81b300312",
    "type": "Preop Pre-anesthesia Evaluation [PDF]",
    "date": "2026-01-30",
    "author": "Brian N. Murphy, MSN, NP-C",

```

```
    "text": "Prior pre-op H&P retained for longitudinal chart context."
  },
  {
    "doc_id": "ffd06a2d-6fc3-5243-9189-6054e87f1c86",
    "type": "Surgical Consent",
    "date": "2026-02-23",
    "author": "Brian O'Connor, DO",
    "text": "Electronic consent obtained and signed by patient for procedure."
  },
  {
    "doc_id": "43d8b80c-0050-5954-a60a-4964b07f0ae3",
    "type": "Pre-op Nursing Intake",
    "date": "2026-02-25",
    "author": "Sarah Murphy, DNP, FNP-BC",
    "text": "Nursing intake confirms home medications and fasting instructions were
reviewed."
  },
  {
    "doc_id": "0e230e4e-469c-503f-9ad7-9ce6d086ab84",
    "type": "Perioperative Medication Plan",
    "date": "2026-02-22",
    "author": "Stephanie Rodriguez, DO",
    "text": "Patient takes apixaban for atrial fibrillation. Follow up with cardiology for
peri-op recommendations."
  }
],
"metadata": {
  "submission_received_at": "2026-03-01T14:22:10Z",
  "source_system": "Cadence Intake API v2"
}
}
```